Dashboard 1: Sales Performance & Music Analytics

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Activity Overview

In this activity, you'll build a complete sales dashboard using the Chinook music store database. You'll create KPI cards, charts, and tables to analyze music sales performance.

What you'll create:

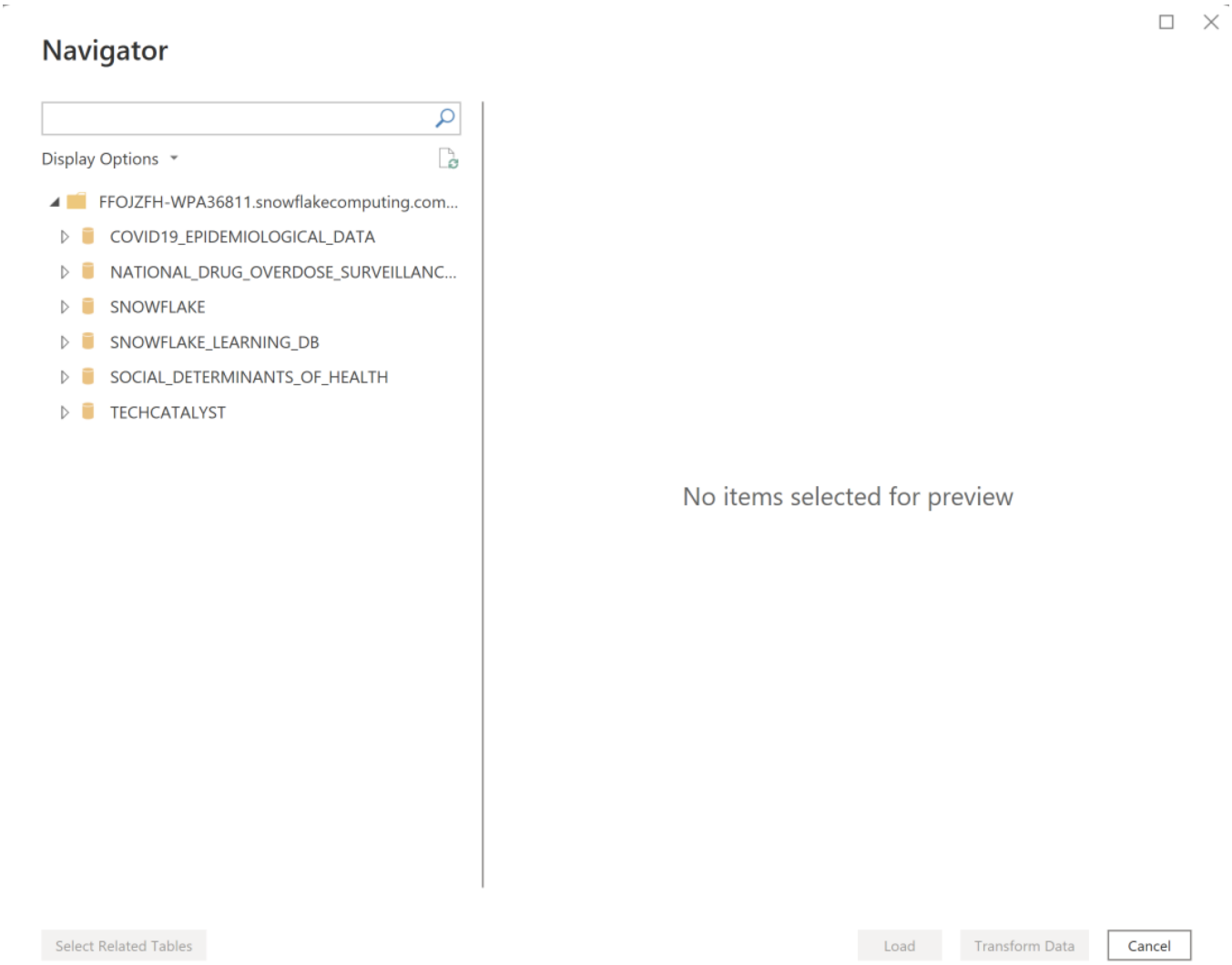
- 4 KPI summary cards
- Revenue trend line chart
- Genre distribution donut chart
- Top artists and albums tables

- Geographic sales map

Step 1: Connect to Snowflake and Load Data

1. **Open Power BI Desktop**
2. **Get Data**:
   - Click **Home** tab → **Get Data** → **More**
   - Search for "Snowflake" and select **Snowflake**
   - Click **Connect**
3. **Enter Connection Details**:
   - Server: `[your-snowflake-account].snowflakecomputing.com`
   - Warehouse: `COMPUTER_WH`
   - Click **OK**
4. **Authenticate**:
   - Enter your Snowflake username and password
   - Click **Connect**
5. **Select Tables**:

   The navigator screen will show.



- In the Navigator window, expand your database and schema
  - Expand the **TECHCATALYST** Database. Then expand the **CHINOOK** schema.
- Check the boxes for these 7 tables:
  - ✅ `INVOICE`
  - ✅ `INVOICE_LINE`
  - ✅ `TRACK`
  - ✅ `ALBUM`
  - ✅ `ARTIST`
  - ✅ `GENRE`
  - ✅ `CUSTOMER`
- Click **Import** (not Transform Data - we'll clean later if needed) then **OK**

6. **Wait for data to load** (you'll see progress in bottom-right corner)

> **Tip**

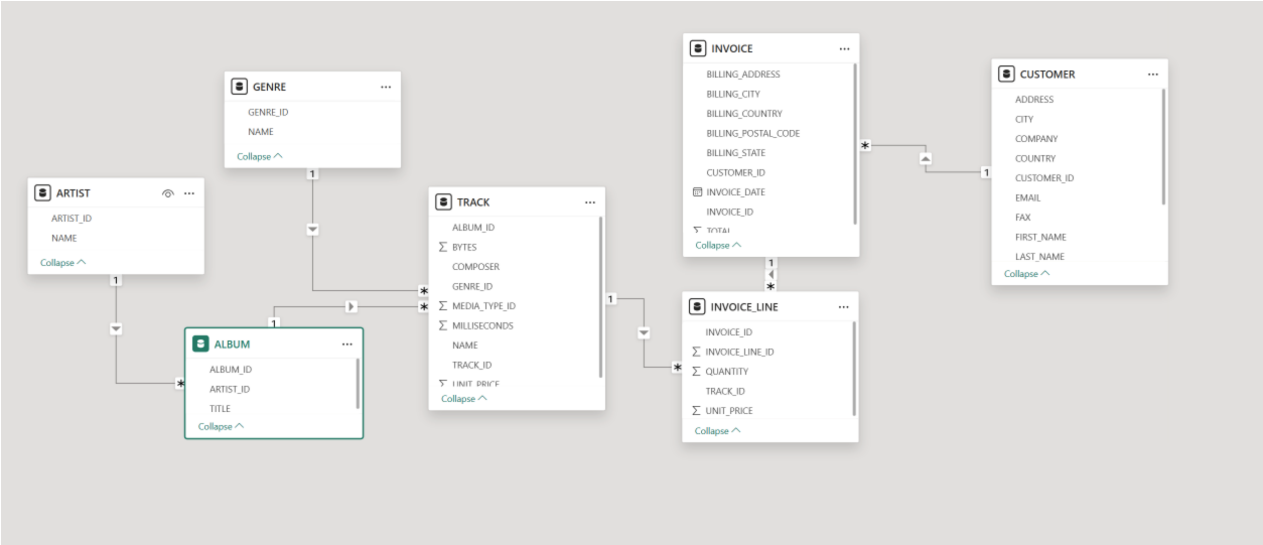
> If tables don't appear, make sure you're looking in the correct Schema. Snowflake uses DATABASE → SCHEMA → TABLE hierarchy.

> **Note**
>
> **Import vs DirectyQuery**
>
> - **Where data lives**
> - **Import**: Data is copied and stored inside the PBIX/Power BI dataset.
> - **DirectQuery**: Data stays in SQL/warehouse/etc.; visuals send queries back to the source.
> - **Performance**
> - **Import**: Generally faster, uses VertiPaq, great for interactive slicing across small–medium data.
> - **DirectQuery**: Typically slower and depends on source DB and network latency.
> - **Freshness**
> - **Import**: Data only updates on scheduled/ manual refresh.

Step 2: Create Relationships in Model View

> **Tip**
>
> If you want to learn about the Chinook data model you can view it here https://github.com/lerocha/chinook-database

1. Click **Model View** icon (left sidebar, bottom icon). **Note that Power BI will attempt to do this for you**. Make sure you confirm the results.
2. **Arrange your tables** - drag them to organize like this:
   - Center: `INVOICE` and `INVOICE_LINE` (fact tables)
   - Around the edges: other tables (dimension tables)
3. **Create the following relationships** (drag from one column to another):
   - `INVOICE[CUSTOMER_ID]` → `CUSTOMER[CUSTOMER_ID]`
   - `INVOICE_LINE[INVOICE_ID]` → `INVOICE[INVOICE_ID]`
   - `INVOICE_LINE[TRACK_ID]` → `TRACK[TRACK_ID]`
   - `TRACK[ALBUM_ID]` → `ALBUM[ALBUM_ID]`
   - `TRACK[GENRE_ID]` → `GENRE[GENRE_ID]`
   - `ALBUM[ARTIST_ID]` → `ARTIST[ARTIST_ID]`



4. **Verify relationships** - each line should show "1" on one side and "*" on the other
5. Click **Report View** to return to canvas

> **Note**
>
> Power BI should auto-detect most relationships. If a relationship already exists, you'll see the line connecting the tables.

Step 3: Create Calculated Columns and Measures

Before building visuals, we need to create some calculations.

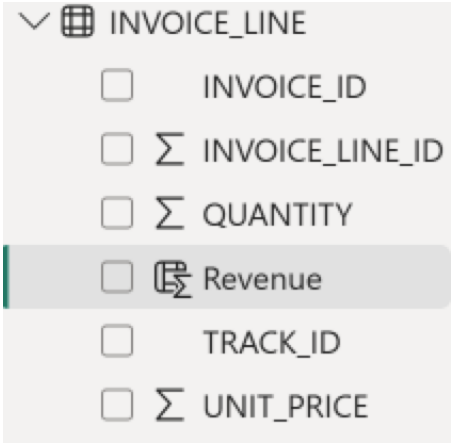
Create Revenue Calculated Column

1. Click on `INVOICE_LINE` table in the **Data** pane
2. **Modeling** tab → **New Column**
3. In the formula bar, type:

```
Revenue = INVOICE_LINE[UNIT_PRICE] * INVOICE_LINE[QUANTITY]
```

4. Press **Enter**

Confirm `Revenue` is created



Create Key Measures

1. **Create Total Revenue Measure:**
   - Right-click on `INVOICE_LINE` table → **New Measure**
   - Type:

```
Total Revenue = SUM(INVOICE_LINE[Revenue])
```

2. **Create Total Invoices Measure:**
   - Right-click on `INVOICE` table → **New Measure**
   - Type:

```
Total Invoices = COUNTROWS(INVOICE)
```

3. **Create Average Order Value Measure:**
   - Right-click on `INVOICE` table → **New Measure**
   - Type:

```
Avg Order Value = DIVIDE([Total Revenue], [Total Invoices])
```

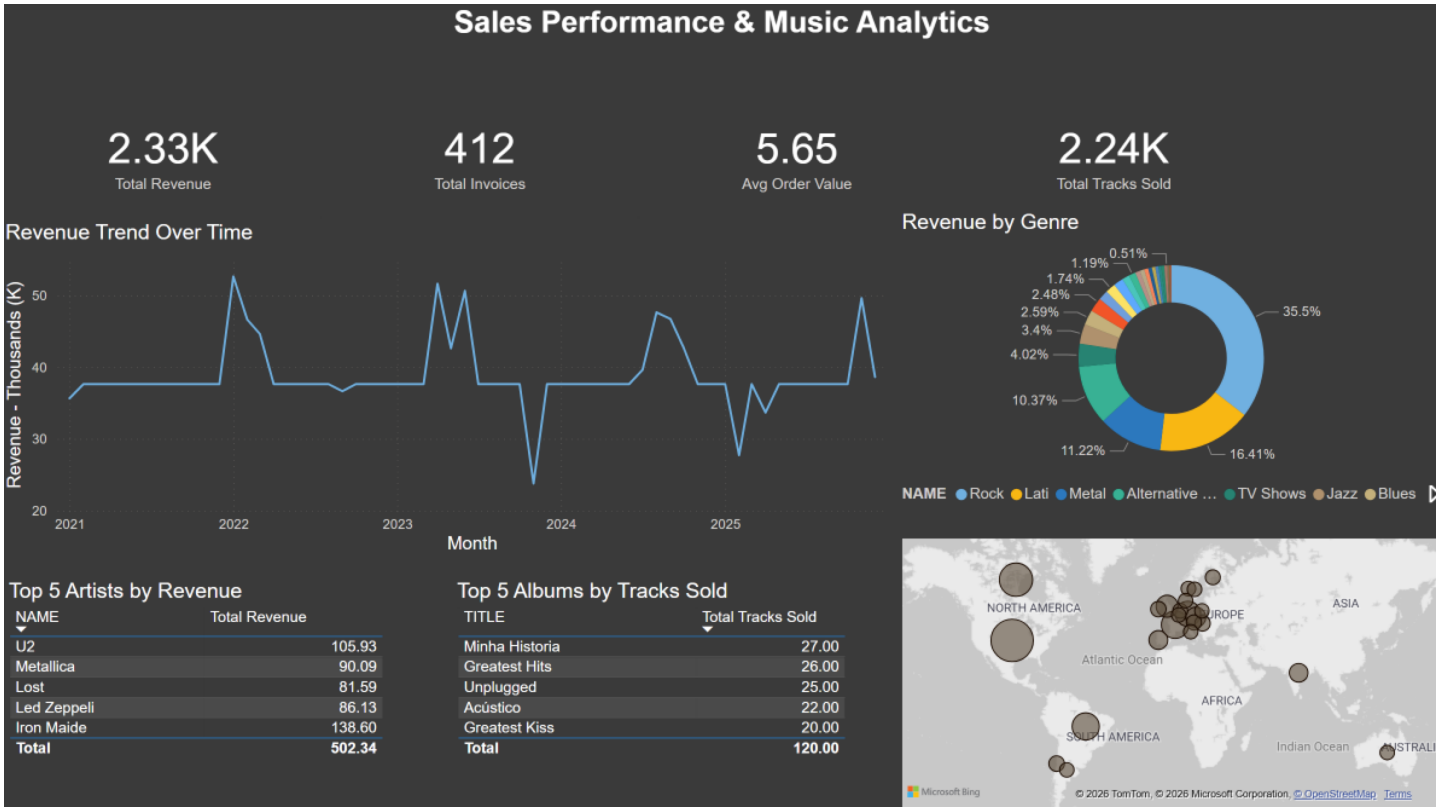
4. **Create Total Tracks Sold Measure:**
   - Right-click on `INVOICE_LINE` table → **New Measure**
   - Type:

```
Total Tracks Sold = SUM(INVOICE_LINE[QUANTITY])
```

> **Important**

> Measures (with Σ icon) are different from columns. Measures calculate dynamically based on context; columns are computed row-by-row.

Step 4: Create KPI Cards



We'll create 4 KPI cards across the top of the dashboard.

KPI Card 1: Total Revenue

1. Click blank area on canvas
2. Select **Card** visual from Visualizations pane
3. Add field:
   - Drag `Total Revenue` measure → **Fields** well
4. Resize and position in top-left corner
5. **Format the card**:
   - Select the card
   - Click **Format visual**
   - **Callout value**:
     - Display units: **Thousands (K)**
     - Font size: **28**

- **Category label:**
  - Text: It is On to display "Total Revenue"
  - Font size: **10**

KPI Card 2: Total Invoices

1. Click blank area
2. Select **Card** visual
3. Drag `Total Invoices` measure → **Fields**
4. Position next to the first card
5. Format similarly (change label to "Total Invoices")

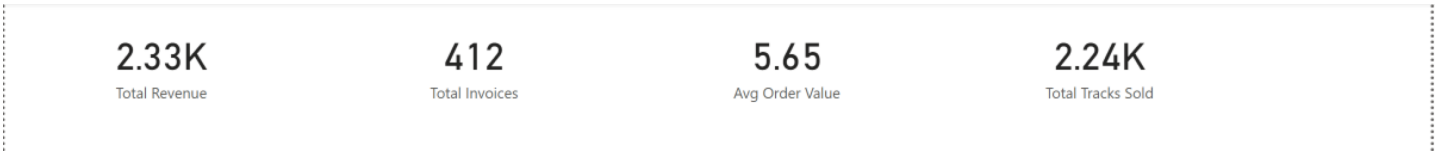
KPI Card 3: Average Order Value

1. Click blank area
2. Select **Card** visual
3. Drag `Avg Order Value` measure → **Fields**
4. Position next to second card
5. Format similarly (label: "Avg Order Value")

KPI Card 4: Total Tracks Sold

1. Click blank area
2. Select **Card** visual
3. Drag `Total Tracks Sold` measure → **Fields**
4. Position next to third card
5. Format similarly (label: "Total Tracks Sold")

> **Tip**
>
> Select all 4 cards (Ctrl+Click), then Format → Align → Align Top to make them uniform.



Step 5: Create Revenue Trend Line Chart

1. Click blank area below the KPI cards

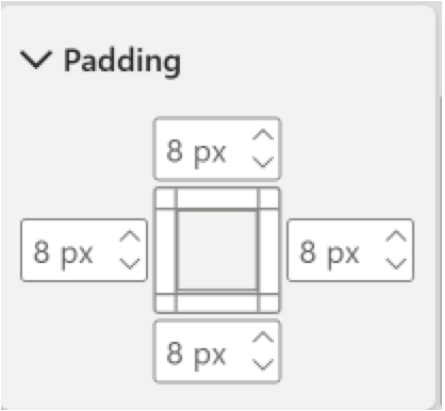
2. Select **Line Chart** from Visualizations pane
3. Add fields:
    - `INVOICE[INVOICE_DATE]` → **X-axis**
    - `Total Revenue` measure → **Y-axis**
    - Click the dropdown on `INVOICE_DATE` in X-axis well
    - Select **Date Hierarchy** → Expand to **Month**
4. Resize to take up about 60% width of the canvas
5. **Format the chart**:
    - **Title**:
        - Title text: "Revenue Trend Over Time"
        - Font size: **14**
    - **X-axis**:
        - Title: "Month"
    - **Y-axis**:
        - Title: “Revenue - Thousands (K)”
    - **Lines**:
        - Stroke width: **2**

Step 6: Create Revenue by Genre Donut Chart

1. Click blank area (to the right of line chart)
2. Select **Donut Chart** from Visualizations pane
3. Add fields:
    - `GENRE[NAME]` → **Legend**
    - `Total Revenue` measure → **Values**
4. Position next to the line chart (remaining 40% width)
5. **Format the chart**:
    - **Title**: "Revenue by Genre"
    - **Legend**: Position **Bottom Center**
    - **Detail labels**: Turn **On** (shows percentages)

Step 7: Create Top 5 Artists Table

1. Click blank area below the charts
2. Select **Table** visual from Visualizations pane
3. Add fields:
    - `ARTIST[NAME]` → **Columns**
    - `Total Revenue` measure → **Columns**
4. **Add Top N Filter**:
    - Select the table
    - In **Filters** pane, find `NAME` under "Filters on this visual"
    - Change filter type to **Top N**
    - Set to show **Top 5**
    - By value: drag `Total Revenue` measure
    - Click **Apply filter**
5. Resize to take up about 1/3 width of row
6. **Format the table**:
    - **Title**: "Top 5 Artists by Revenue"
    - **Style**: **Minimal**
    - **Grid**:
        - padding: **8**



Step 8: Create Top 5 Albums Table

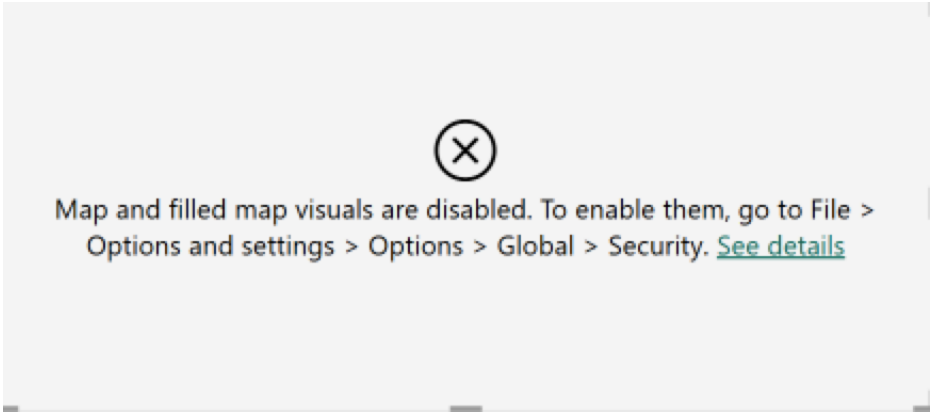
1. Click blank area next to artists table
2. Select **Table** visual
3. Add fields:
    - `ALBUM[TITLE]` → **Columns**

    - `Total Tracks Sold` measure → **Columns**

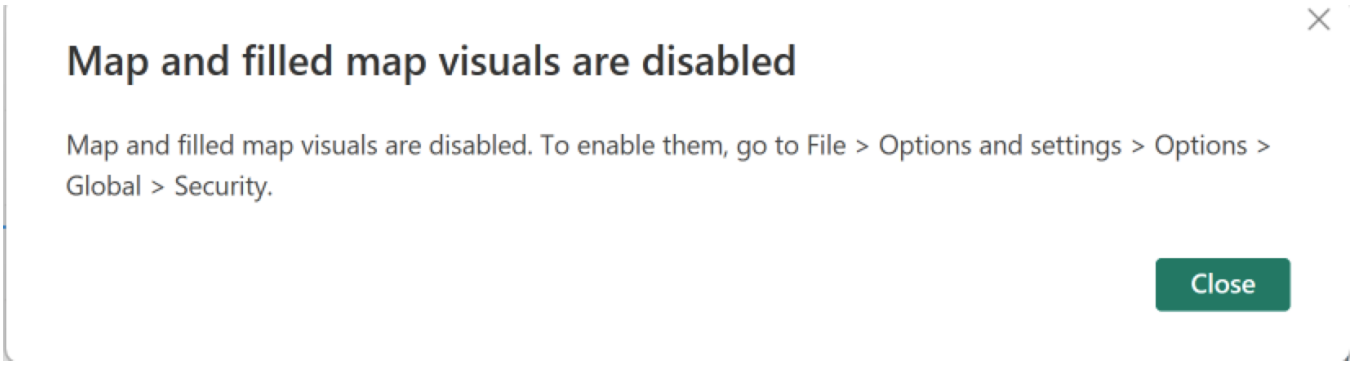
4. **Add Top N Filter**:
    - Filter `TITLE` to **Top 5**
    - By value: `Total Tracks Sold`
    - Apply filter
5. Resize to take up about 1/3 width
6. **Format the table**:
    - **Title**: "Top 5 Albums by Tracks Sold"
    - Format similarly to artists table

Step 9: Create Sales by Country Map

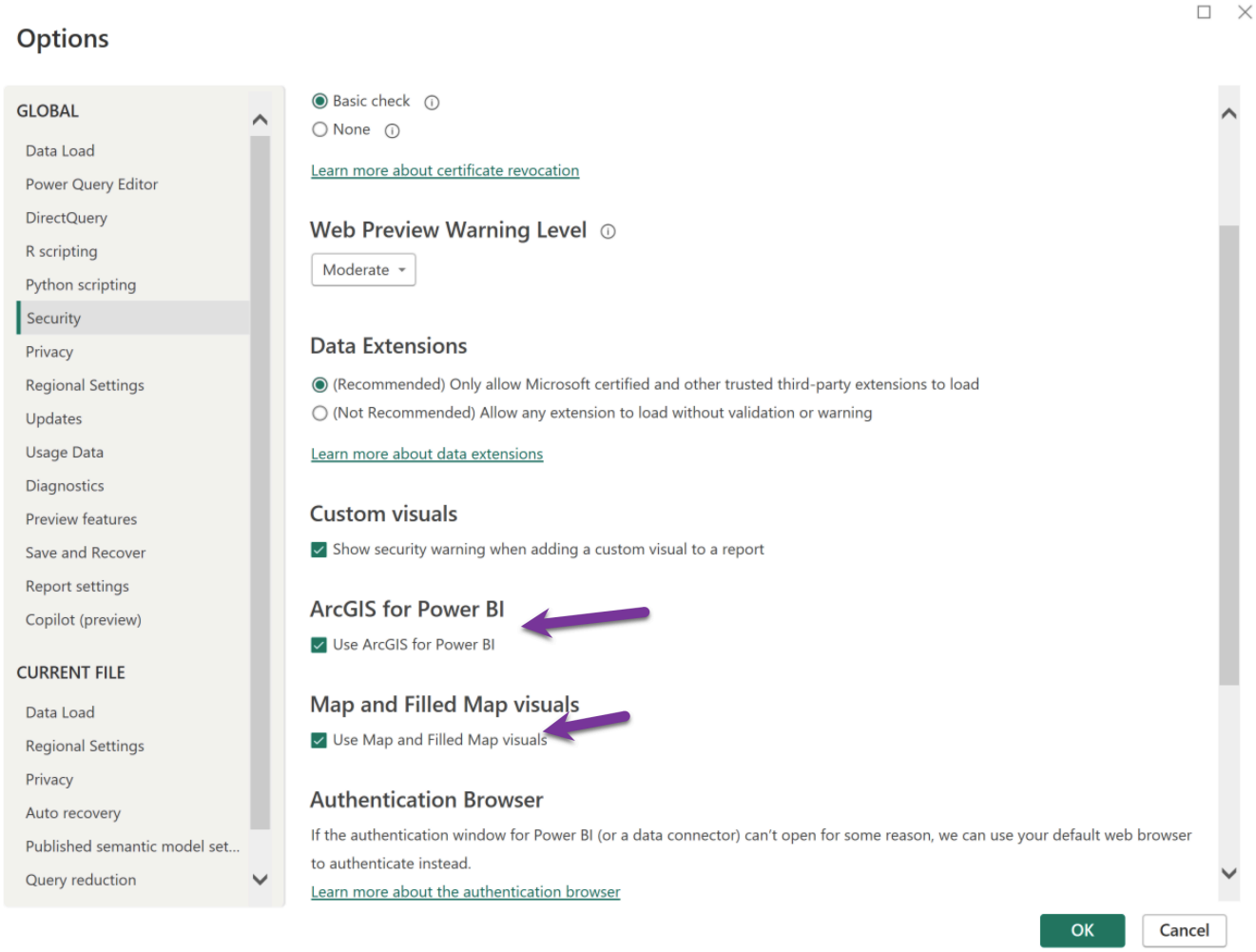
1. Click blank area next to albums table
2. Select **Map** visual (or **Filled Map**) from Visualizations pane



If you click See Details you will get



Follow the steps



Sometimes you will need to close the application and re-open for it to take effect. Make sure to save your work first.

3. Add fields:
   - `CUSTOMER[COUNTRY]` → **Location**
   - `Total Revenue` measure → **Bubble Size** (for Map) or **Color saturation** (for Filled Map)
4. Resize to remaining 1/3 width
5. **Format the map**:
   - **Title**: "Sales by Country"
   - **Data colors**: Choose a color gradient
   - **Zoom buttons**: Turn **On**

> **Note**
>
> If countries don't appear correctly, Power BI may need to geocode them. Click the warning icon and select "Yes, use as geographic data."

Step 10: Final Formatting and Polish

1. **Add a report title**:
    - **Insert** tab → **Text box**
    - Type: "Sales Performance & Music Analytics"
    - Format: Font size **20**, Bold, Center align
    - Position at very top of page
2. **Apply consistent colors**:
    - From View select a theme.
3. **Test interactions**:
    - Click on different segments in the donut chart
    - Verify all visuals filter appropriately

Step 11: Save Your Work

1. **File** → **Save As**
2. Name: `Chinook_Sales_Dashboard.pbix`
3. Click **Save**

Expected Results

When complete, your dashboard should show:

- 4 KPI cards at the top showing totals
- A line chart showing revenue trends by month
- A donut chart showing revenue distribution by genre
- A table of top 5 artists by revenue
- A table of top 5 albums by tracks sold
- A geographic map showing sales by country

Total visuals: 9 (4 cards + 5 charts/tables)

Common Issues and Solutions

Issue: Relationships Won't Create

Solution: Make sure column data types match. Both columns must be INT or both must be TEXT. Check in Table View.

Issue: Measures Show Wrong Totals

Solution: Don't use calculated columns in measures. Use `SUM(INVOICE_LINE[Revenue])` not `SUM(Revenue)` without table name.

Issue: Map Doesn't Show Countries

Solution: Click the warning icon on the visual and confirm you want Power BI to use the field as geographic data.

Issue: Dates Don't Sort Correctly

Solution: Make sure INVOICE_DATE is recognized as Date type. Check in Model View → Column properties.

Issue: Visuals Are Blank

Solution: Check filters pane. You may have accidentally filtered out all data. Clear filters and try again.

Bonus Challenges

If you finish early, try these enhancements:

1. **Add a slicer** for Genre - let users filter the entire dashboard by music genre
2. **Create a second page** showing customer analytics (most valuable customers, customers by country)
3. **Add conditional formatting** to the Top 5 tables - highlight the #1 artist/album
4. **Create a YTD (Year-to-Date) revenue measure** using DAX time intelligence
5. **Add drill-through** from the map to a detail page showing individual customer purchases

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What You Learned

✅ How to connect Power BI to Snowflake ✅ Creating star schema relationships in Model View ✅ Building DAX measures for aggregations ✅ Creating KPI cards for dashboard summaries ✅ Using Top N filters on visuals ✅ Formatting and polishing a professional dashboard

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